

Chapter 2

1

Introduction to Probability

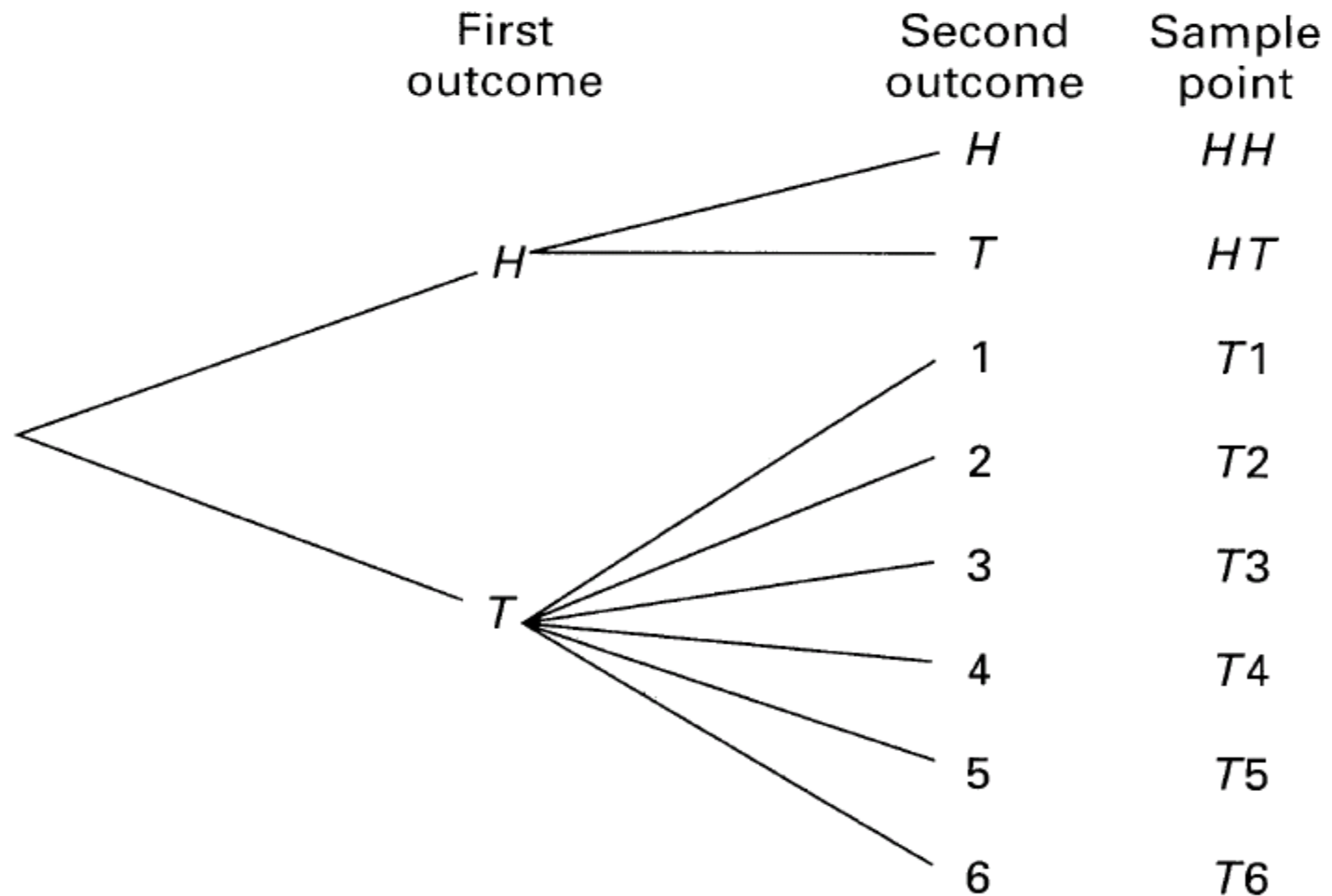
2.1 Sample Space

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- ***Sample Space***: The set of all possible outcomes of a statistical experiment
- ***Sample Point or Element***: Each outcome in a sample space
- *See Examples 2.1, 2.2, 2.3*

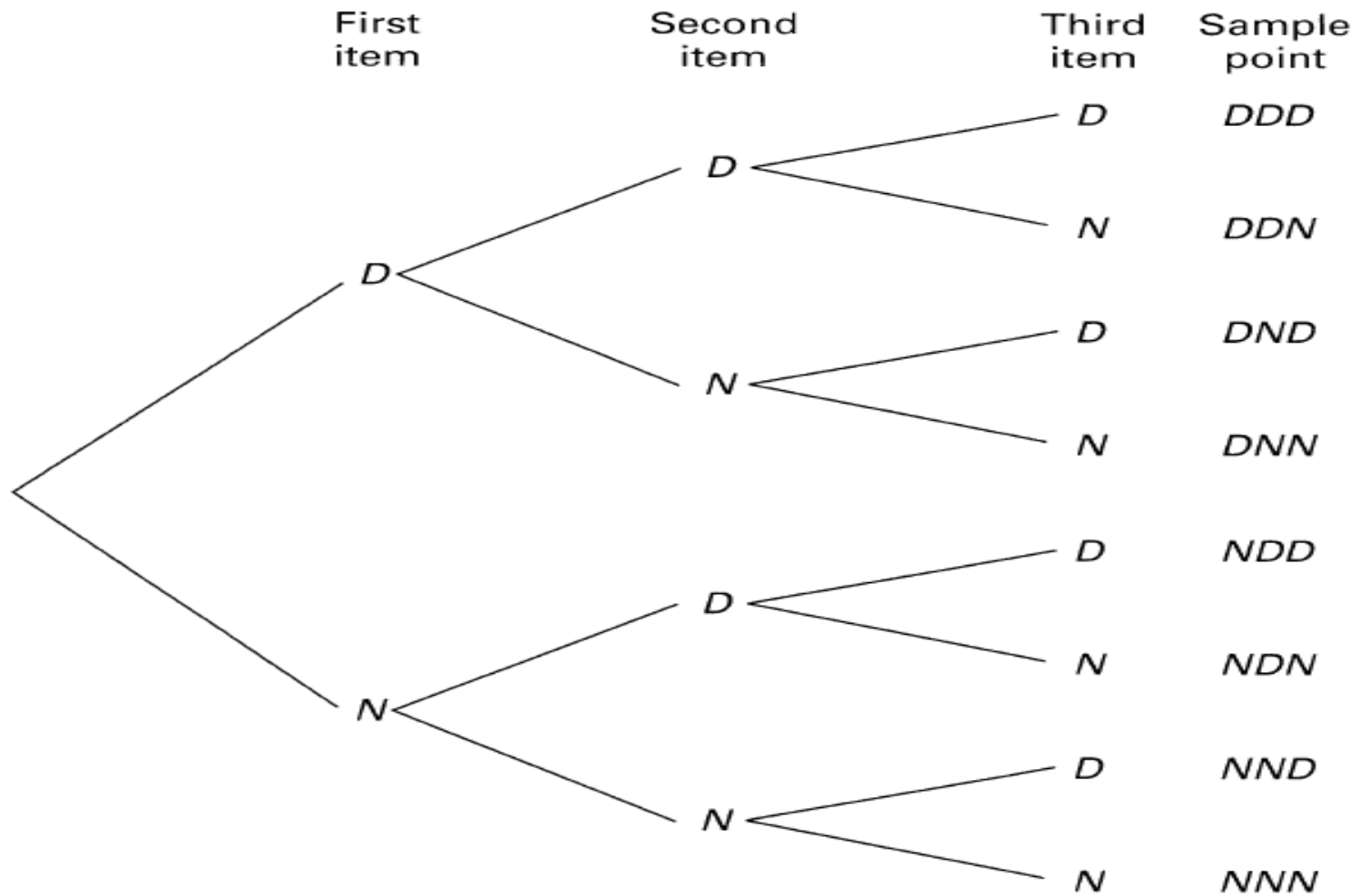
Example 2.2

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Example 2.3

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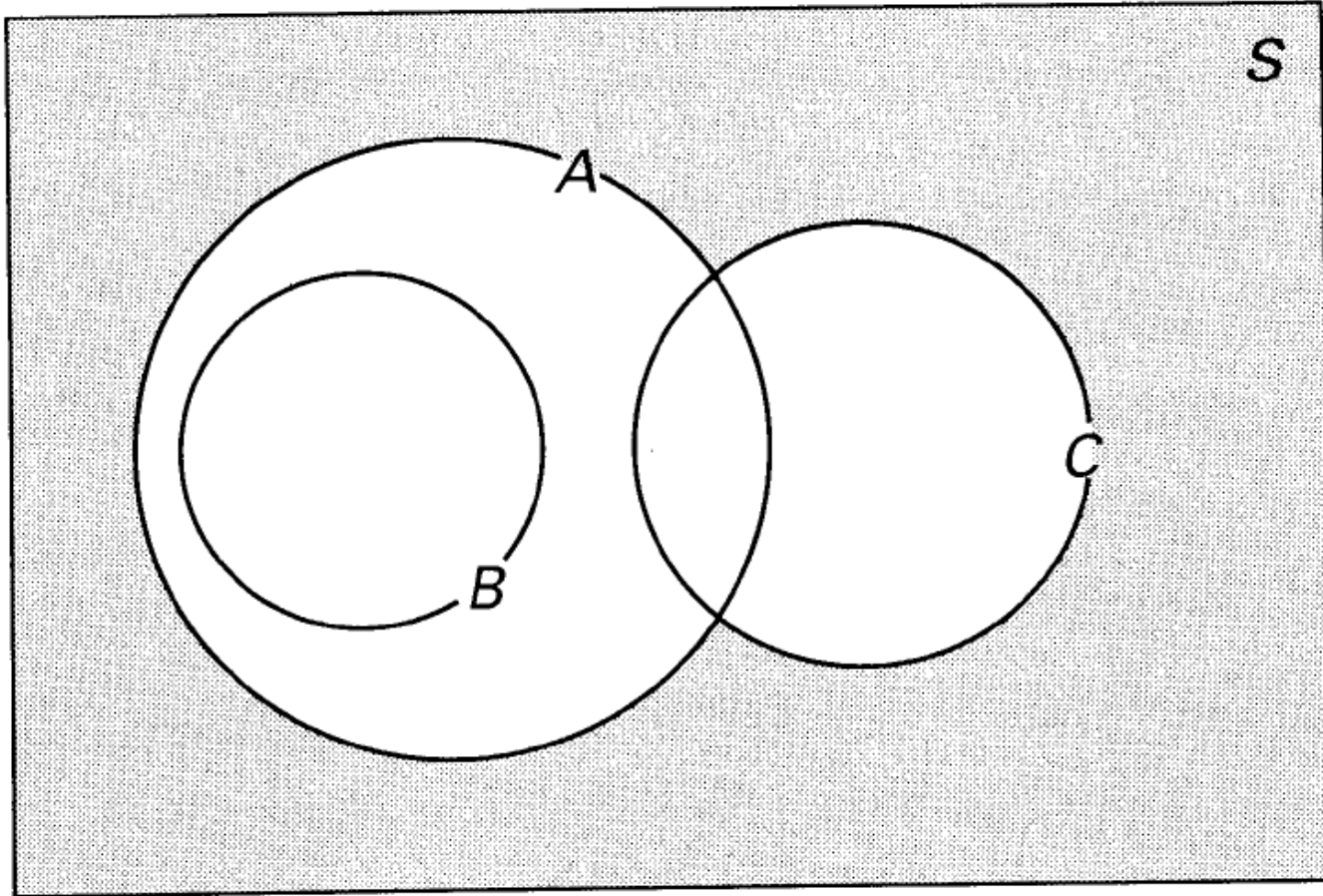
2.2 Event

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- **Event:** A subset of a sample space S
- **Null Set** (ϕ): An event contains no elements
- **Complement** of an event A (A'): The event of all elements of S that are not in A
- **Intersection** of two events A and B ($A \cap B$): The event contains all elements that are common to A and B
 - If $A \cap B = \emptyset$ then events A and B are *mutually exclusive* (i.e., A and B cannot occur simultaneously)
- **Union** of two events A and B ($A \cup B$): The event contains all elements that belong to A or B or both
- **Exercises: 2.7, 2.14, 2.16**

Venn Diagram

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2.3 Counting Sample Points

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- ***Multiplication Rule:*** If an operation can be performed in n_1 ways, and if for each of these a second operation can be performed in n_2 ways, and so forth, then the sequence of k operations can be performed together in $n_1 n_2 \dots n_k$ ways
- *See Examples 2.13, 2.17*

2.3 Counting Sample Points

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- ***Permutation***: An arrangement of a set of objects
 - The number of permutations of n distinct objects is $n!$
 - The number of permutations of n distinct objects arranged in a circle is $(n-1)!$
 - The number of permutations of n distinct objects taken r at a time is

$${}_nP_r = n(n-1)(n-2)\cdots(n-r+1) = \frac{n!}{(n-r)!}$$

- See Example 2.19

2.3 Counting Sample Points

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- ***Combination***: A selection from a set of objects
 - The number of combinations of n distinct objects taken r at a time is

$$\binom{n}{r} = \frac{n!}{r!(n-r)!}$$

- See *Example 2.23*

2.3 Counting Sample Points

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- **Partition:** The way of partitioning a set of objects
 - The number of ways of partitioning n distinct objects into r cells with n_1 elements in cell 1, n_2 elements in cell 2, and so forth, is

$$\binom{n}{n_1, n_2, \dots, n_r} = \frac{n!}{n_1!n_2!\cdots n_r!}$$

- See Example 2.21
- **Exercises:** 2.33, 2.34, 2.45

2.4 Probability of an Event

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- ***Probability of an event A :*** The sum of probabilities of all sample points in A , denoted by $P(A)$

$$0 \leq P(A) \leq 1, \quad P(\phi) = 0, \quad \text{and} \quad P(S) = 1$$

- *See Examples 2.24, 2.25, 2.28*

2.5 Additive Rules

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- **Additive Rule:** If A and B are any two events, then

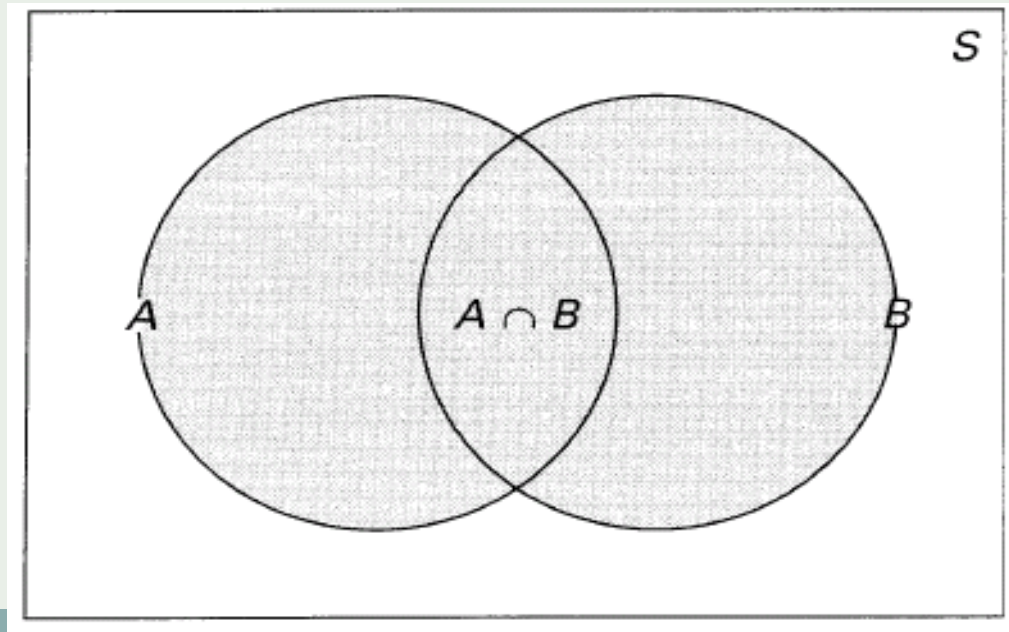
$$P(A \cup B) = P(A) + P(B) - P(A \cap B)$$

- If $A_1, A_2, A_3, \dots$, and A_k are mutually exclusive, then

$$P(A_1 \cup A_2 \cup A_3 \cup \dots \cup A_k) = P(A_1) + P(A_2) + P(A_3) + \dots + P(A_k)$$

- See Examples 2.29, 2.30

- **Exercises:** 2.49, 2.59



2.6 Conditional Probability

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- **Conditional Probability** $P(B|A)$: The probability that an event B occurs given that the event A occurs

$$P(B|A) = \frac{P(A \cap B)}{P(A)} \quad \text{if} \quad P(A) > 0$$

- Two events A and B are *independent* if and only if

$$P(B|A) = P(B) \quad \text{or} \quad P(A|B) = P(A)$$

- See Example 2.34

2.6 Conditional Probability

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- **Multiplicative Rule:** If events A and B can both occur, then

$$P(A \cap B) = P(A)P(B|A)$$

- Two events A and B are *independent* if and only if

$$P(A \cap B) = P(A)P(B)$$

- If events $A_1, A_2, A_3, \dots$, and A_k can occur, then

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_k)$$

$$= P(A_1)P(A_2|A_1)P(A_3|A_1 \cap A_2) \dots P(A_k|A_1 \cap A_2 \cap \dots \cap A_{k-1})$$

- See Examples 2.37, 2.39, 2.40
- **Exercises:** 2.80, 2.83, 2.92, 2.93

2.6 Mutually Exclusive Versus Independent

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- **Mutually Exclusive** events are those cannot occur simultaneously

$$P(A \cap B) = 0$$

- **Independent** events are those for which the occurrence of some does not affect

$$P(A \cap B) = P(A)P(B)$$

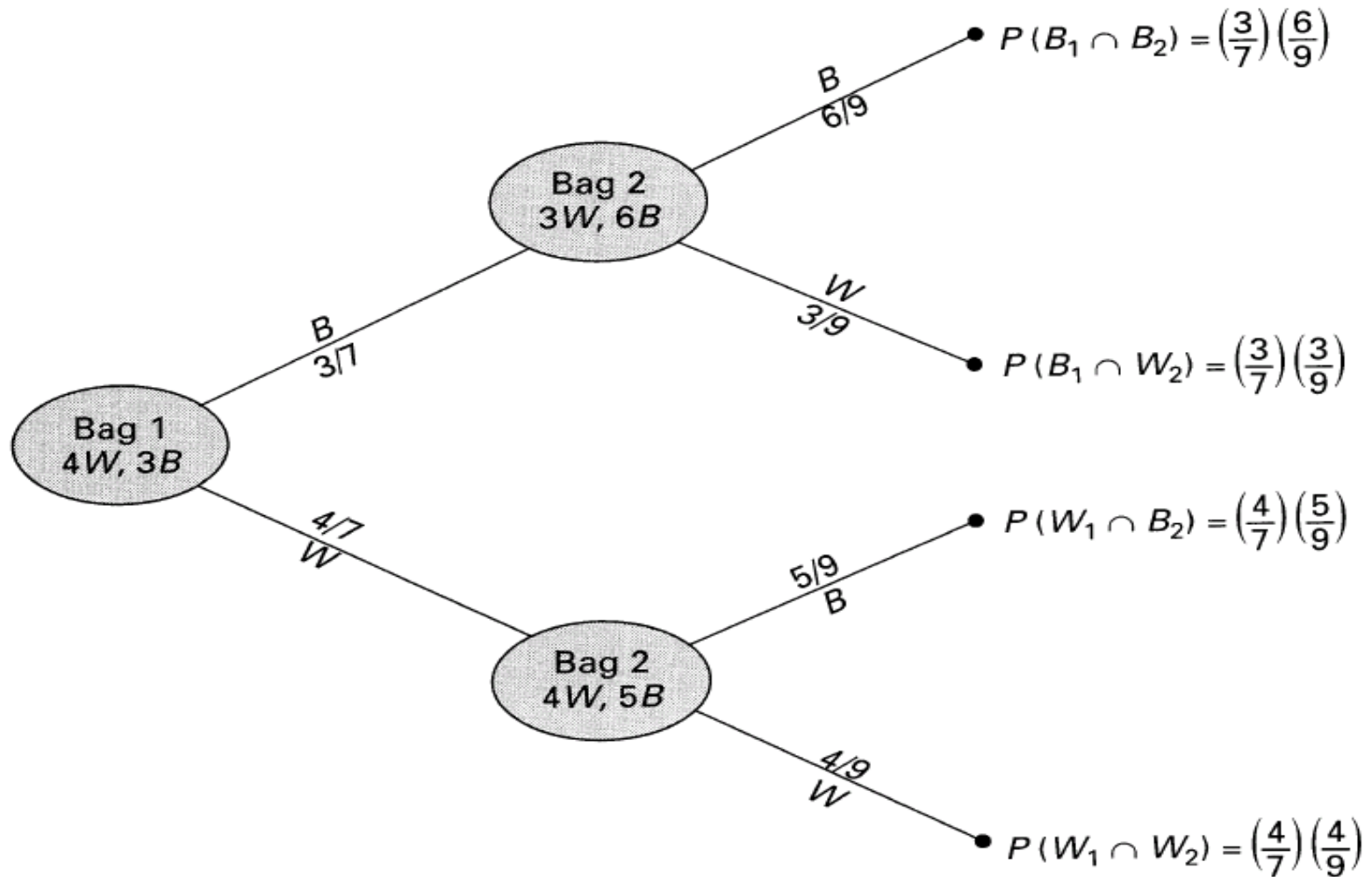
- If events $A_1, A_2, A_3, \dots$, and A_k can occur, then

$$P(A_1 \cap A_2 \cap A_3 \cap \dots \cap A_k)$$

$$= P(A_1)P(A_2 | A_1)P(A_3 | A_1 \cap A_2) \dots P(A_k | A_1 \cap A_2 \cap \dots \cap A_{k-1})$$

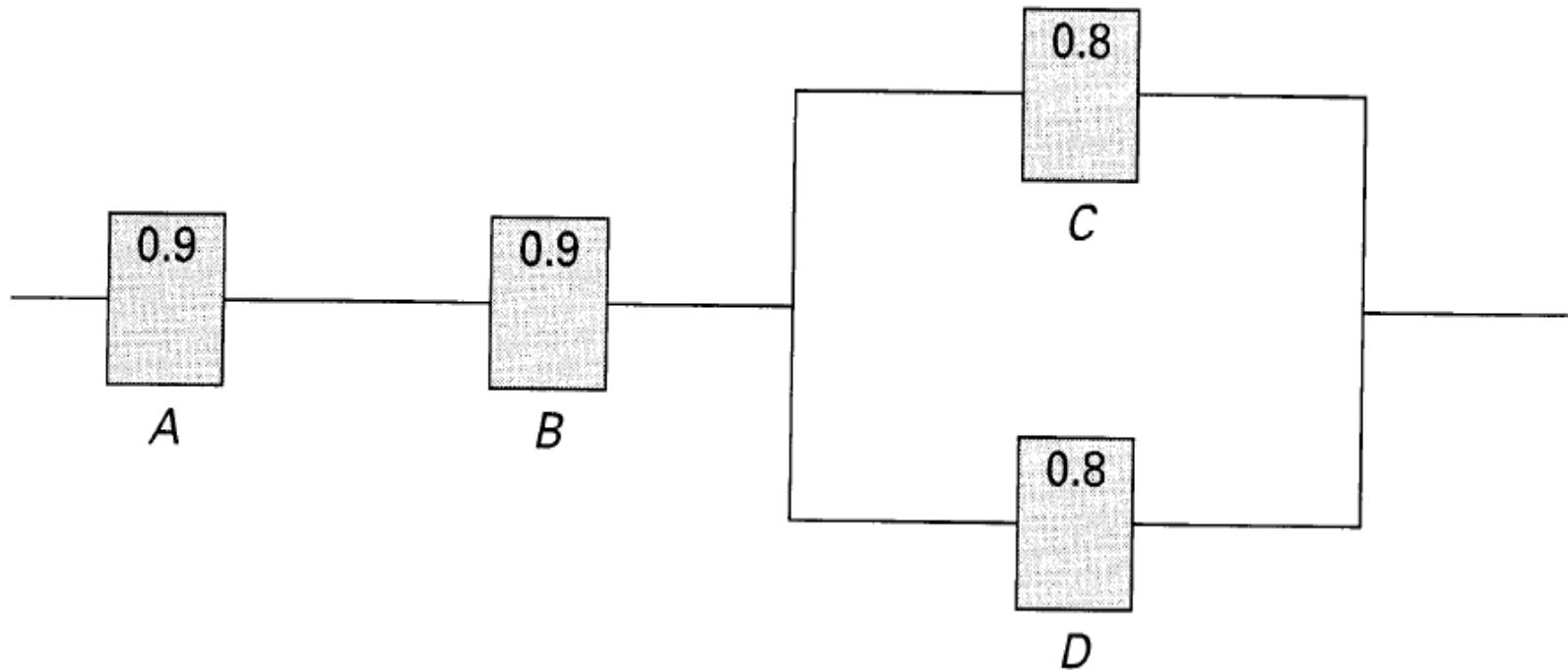
Example 2.37

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Example 2.39

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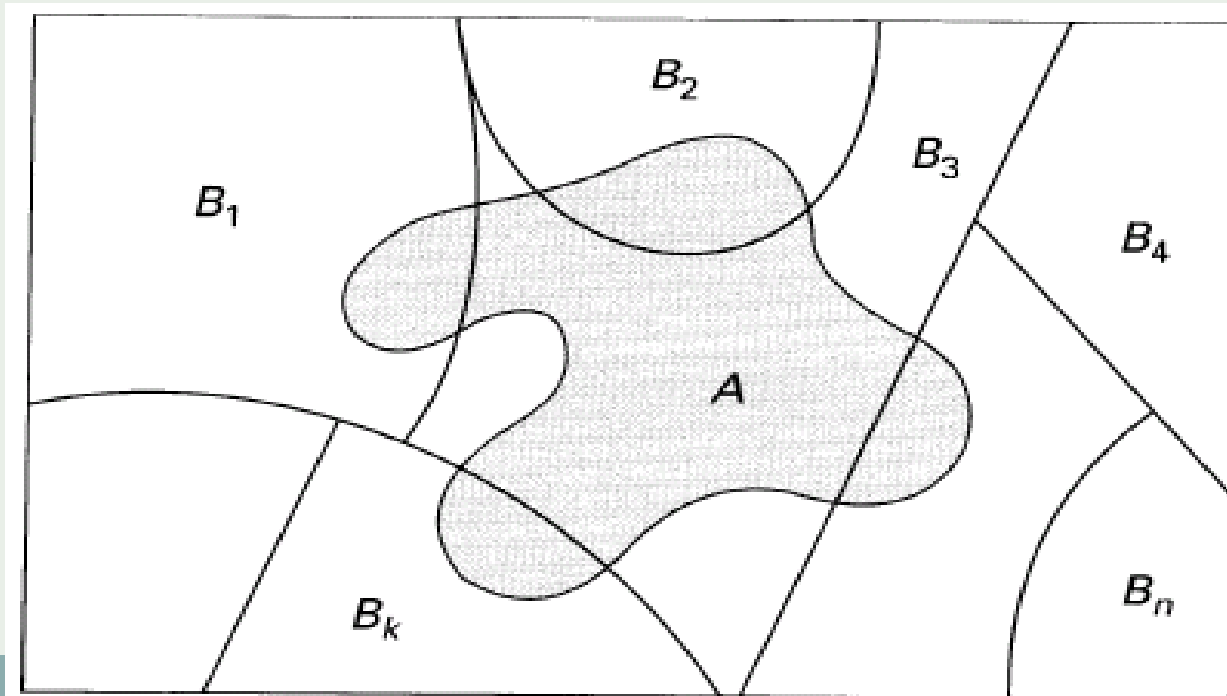


2.7 Bayes' Rule

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- If events $B_1, B_2, \dots$, and B_k constitute a partition of the sample space S , then for any event A in S

$$P(A) = \sum_{i=1}^k P(B_i \cap A) = \sum_{i=1}^k P(B_i)P(A|B_i)$$



2.7 Bayes' Rule

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- **Bayes' Rule:** If events $B_1, B_2, \dots$, and B_k constitute a partition of the sample space S , then for any event A in S

$$P(B_r | A) = \frac{P(B_r \cap A)}{\sum_{i=1}^k P(B_i \cap A)} = \frac{P(B_r)P(A|B_r)}{\sum_{i=1}^k P(B_i)P(A|B_i)}$$

- See Examples 2.41, 2.42
- **Exercises:** 2.96, 2.98, 2.119, 2.120, 2.124, 2.125

Example 2.41

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